

GRADUATE LINEAR ALGEBRA SYLLABUS

COURSE DESCRIPTION

Abstract vector spaces and linear transformations, eigenvectors and eigenvalues, inner product spaces, matrix decompositions, normed spaces, duality, bilinear forms, multilinear algebra.

WEEKLY TOPICS

Week 1 **Review.** (*All proofs are included in the course notes but they are not covered in class during this week.*)

Vector space structure of $\mathbb{R}^n$. Precise definitions of subspace, span, linear independence, basis, dimension. Matrices and linear transformations: representing linear transformation by a matrix, effect of change of basis on matrix representation, kernel and range as subspaces, similarity. Determine spanning sets for kernel and range when given a matrix. Vector space isomorphisms.

Weeks 2-3 **Abstract vector spaces and linear transformations.**

Direct sums and quotient spaces. Trace and determinant, including their properties such as being independent of basis.

Week 4 **Eigenvalues and eigenvectors.**

Eigenvalues and eigenvectors, characteristic polynomials, Cayley–Hamilton theorem, diagonalization.

Weeks 5-6 **Inner product spaces.**

Inner product spaces over $\mathbb{R}$, in coordinates and basis-free. Orthogonality, orthogonal bases, Gram-Schmidt process, orthogonal projections, orthogonal complement. Adjoint, self-adjoint, orthogonal operators (for matrices: transpose, symmetric, and orthogonal). Spectral theorem for self-adjoint operators (in coordinates: symmetric matrices). Complex inner product (unitary) spaces, orthogonality in them, change of basis via unitary operators, adjoint operator (for matrices: conjugate transpose). Spectral theorem for Hermitian operators (in coordinates: Hermitian matrices).

Weeks 7-8 **Matrix decompositions.**

Jordan canonical form, positive definite matrices, QR, SVD, polar decomposition. Some of these rely on inner products and orthogonality in weeks 5-6.

Week 9 **Norms on vector spaces and linear transformations.**

Vector space norms, equivalence of norms in finite-dimensional $\mathbb{R}$ -vector spaces, matrix and operator norms, Frobenius norm, operator norm formula in terms of the maximum eigenvalue of AA^T .

Weeks 10-11 **Dual spaces.**

Linear functionals, dual space. Duality with inner products. Duality of vector

spaces, bases, and linear maps. Relate dual linear map and transpose matrix. Double duality.

Week 12 **Bilinear forms.**

Degenerate and nondegenerate bilinear forms, their matrix representation, change of basis. Quadratic forms: relation to symmetric bilinear forms, equivalence, diagonalization, positive-definiteness, signature.

Weeks 13-14 **Multilinear algebra.**

Tensor products and exterior powers of vector spaces and linear maps between them, both basis-free and in matrix form. Tensor and exterior algebras of a vector space.

POSSIBLE APPLICATION TOPICS (EXTEND THIS LIST OVER TIME)

- Perron-Frobenius theorem (week 9)
- Least squares using QR or SVD
- Graph theory, e.g. Kirchhoff's matrix tree theorem
- Connections to random walks and Markov chains
- Minimax theory
- Gershgorin discs
- Applications in differential equations (e.g., exponentiating matrices to solve systems)

LECTURE NOTES AND REFERENCE BOOKS

Keith and Vasilis will prepare a set of lecture notes which will be available to all faculty. Some related books:

- (1) Lax: Linear Algebra and Its Applications
- (2) Hoffman and Kunze: Linear Algebra
- (3) Johnson and Horn: Matrix Analysis